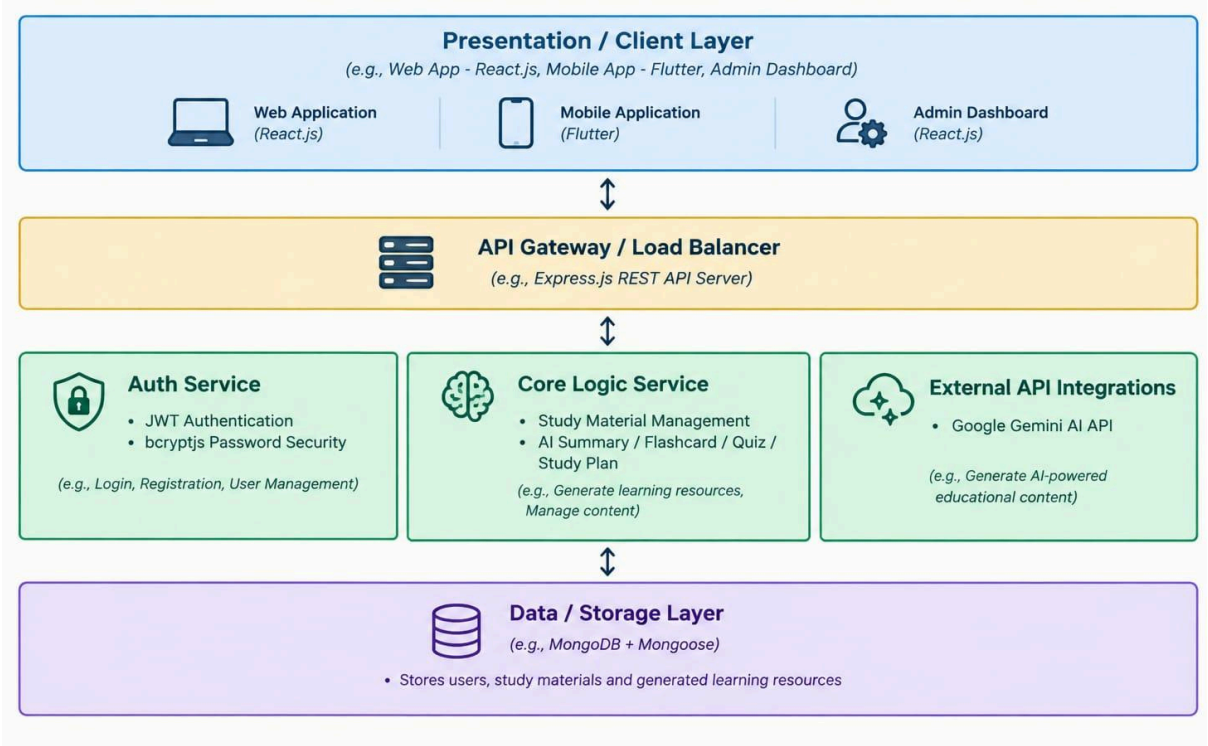


SOLUTION ARCHITECTURE

Date	28 September 2026
Team ID	SWTID-2026-6259
Project Name	AI StudyBuddy
Maximum Marks	5 Marks

Solution Architecture Diagram:



Component Description Table:

Component Name	Description/Role in Architecture	Technological used
Presentation/ client layer	Provides the user interface for students and administrators.	React.js

API Gateway/ server	Receives requests from the frontend and routes them the to the appropriate backend services.	Node.js,Express.js
Backend Services	Handles authentication, study materials, AI content generation and business logic.	JWT,bcryptjs, Google Gemini AI API
Data/Storage Layer	Stores user data,study materials,and AI-generated learning resources securely.	MongoDB, Mongoose